

Introduction

This project develops an AI-assisted SOC pipeline for detecting DoS attacks using machine learning. Random Forest and Isolation Forest were compared, and the selected model was integrated with risk scoring, explainability, MITRE ATT&CK mapping, human review, logging, and monitoring.

Methodology & System Architecture

Dataset: CICIDS2017 - Wednesday Working Hours

Task: Binary classification - Normal vs. DoS

Models: Random Forest and Isolation Forest

Train/Test Split: 80/20 stratified

Selected Features: 12 network flow features

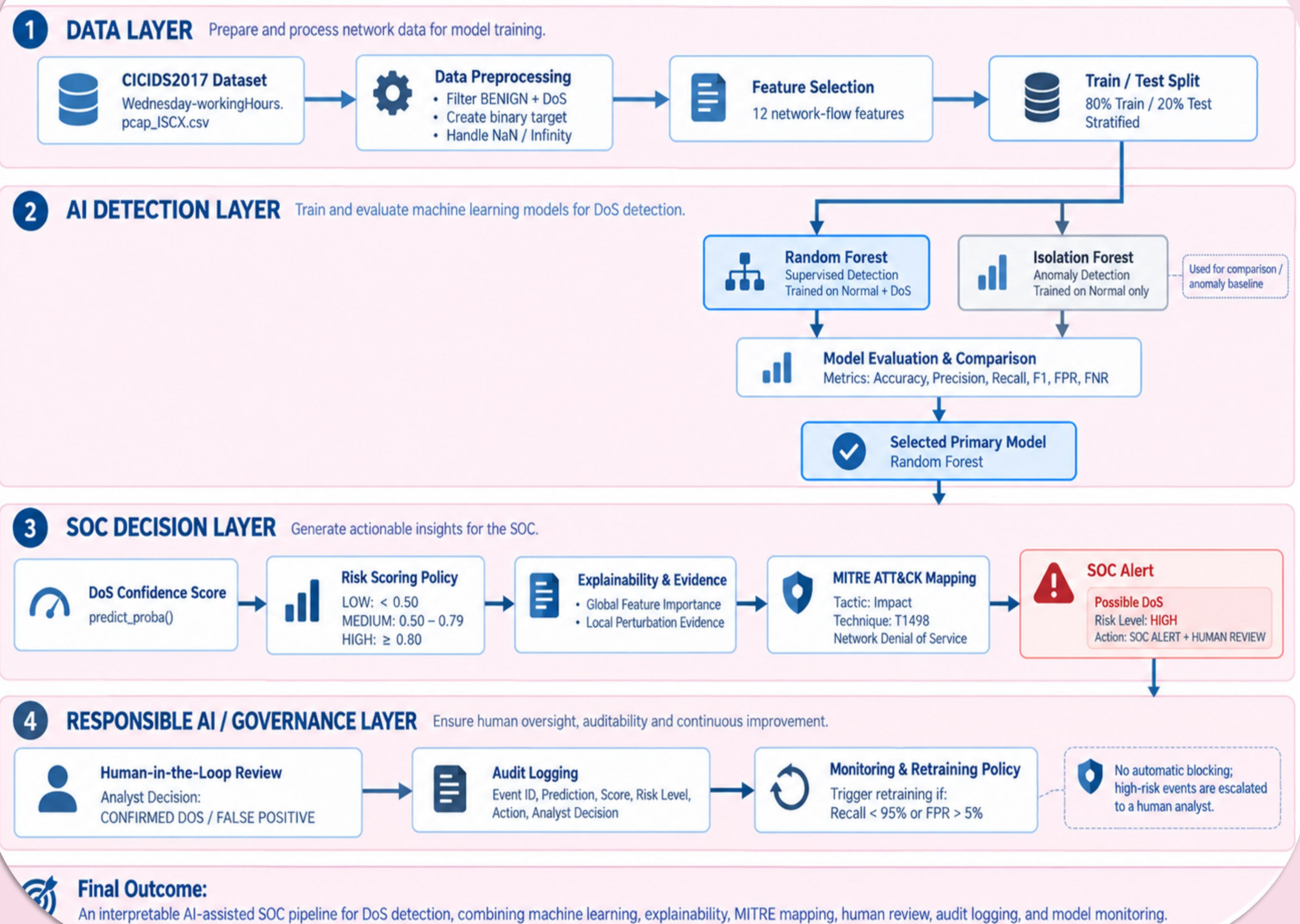
Source Code / Project Repository



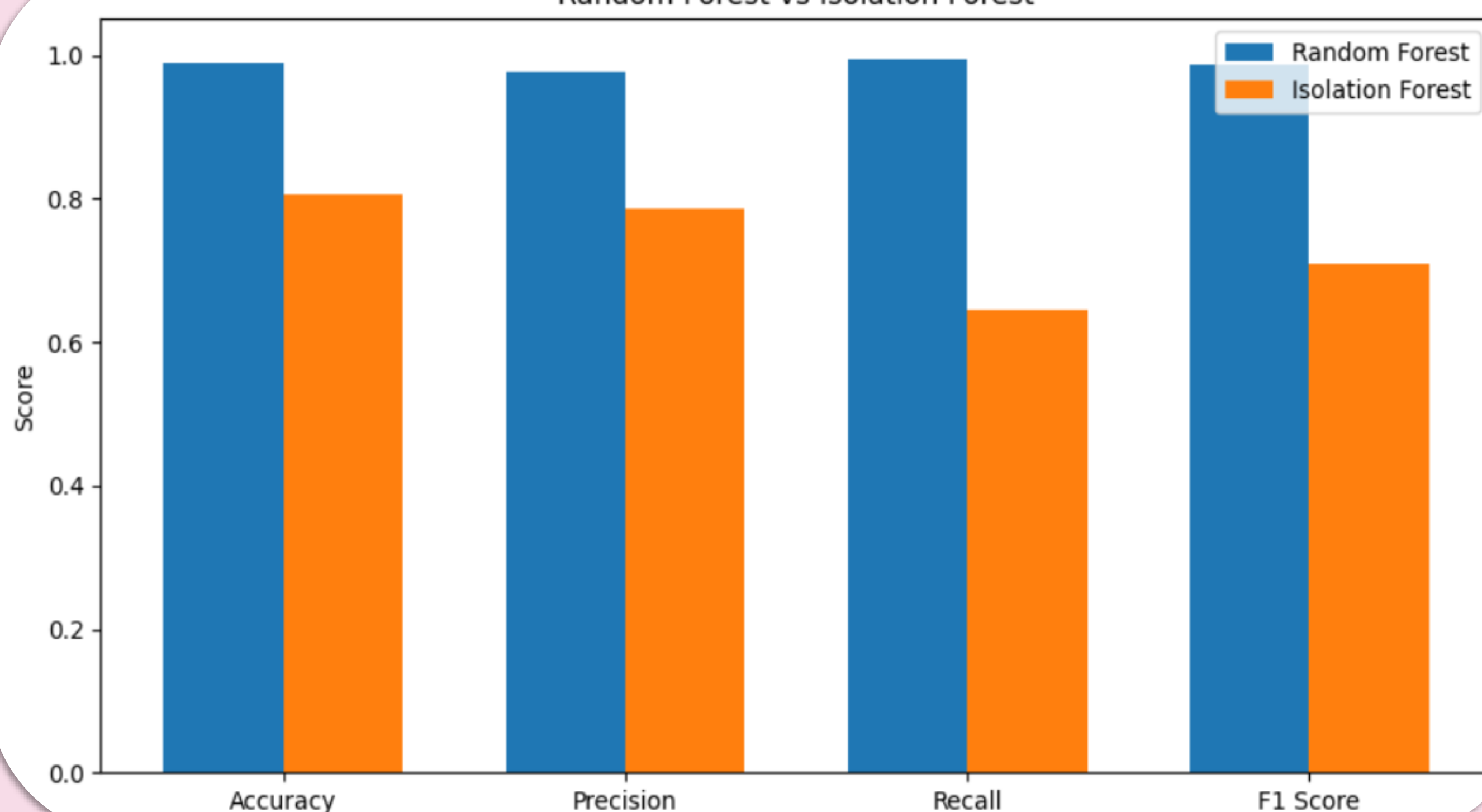
GitHub Repository
Notebook + SOC log + source files

AI-Based DoS Detection for SOC

Architecture and End-to-End Workflow



Random Forest vs Isolation Forest



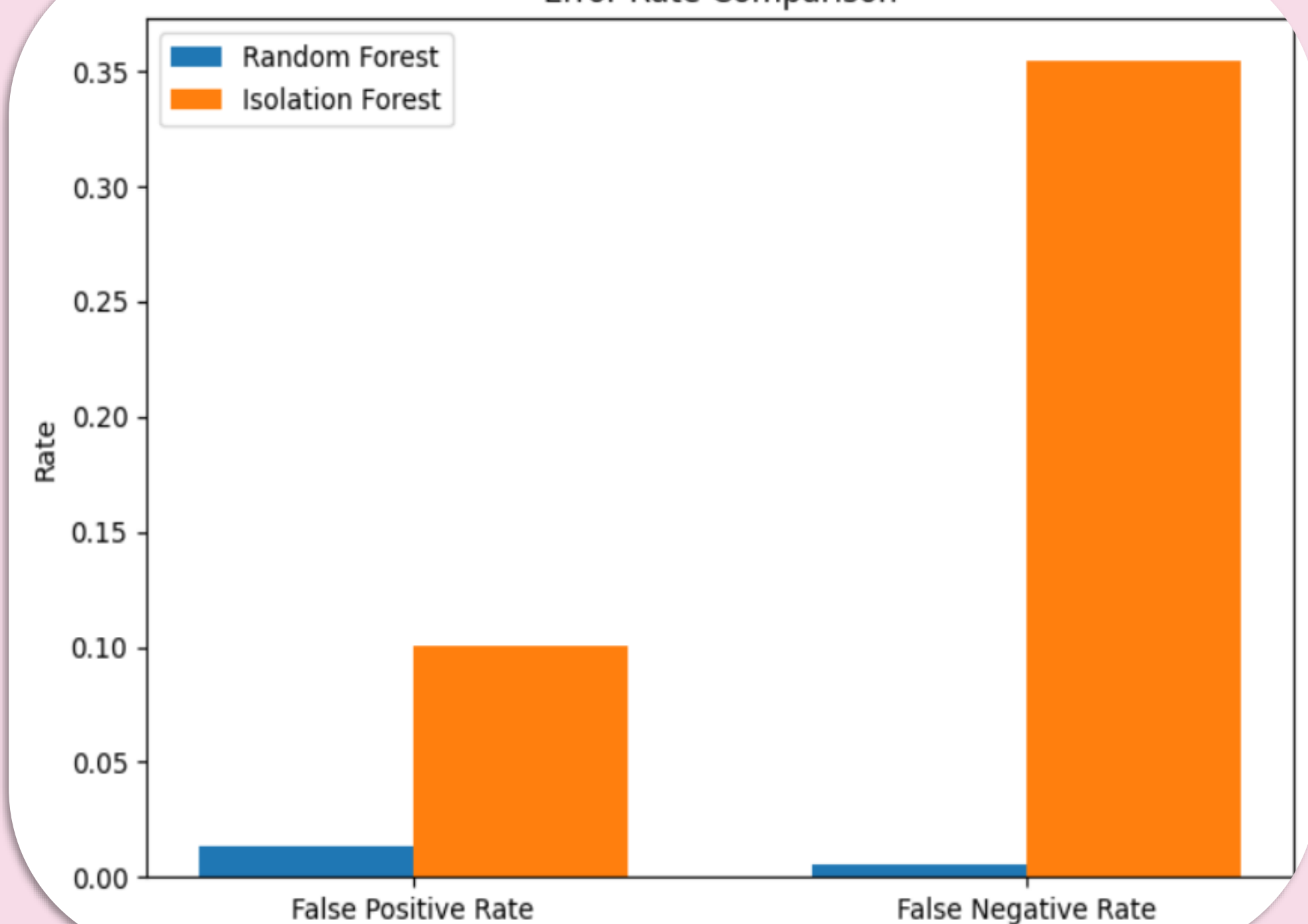
Conclusions

- Random Forest significantly outperformed Isolation Forest.
- The final pipeline provides risk scoring, explainability and MITRE ATT&CK context.
- Human review prevents fully automated security decisions.

Results

Random Forest achieved consistently better performance, with 98.96% Accuracy and 99.45% Recall, and was selected as the primary DoS detection model.

Error Rate Comparison



Discussions / Future Work

- Evaluate the model on additional and real-time network traffic.
- Extend the system with additional attack categories, drift monitoring and SIEM integration.